ASSIGNMENT- 10.5

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Batch: 20

Task Description #1 – Variable Naming Issues Sample

input :

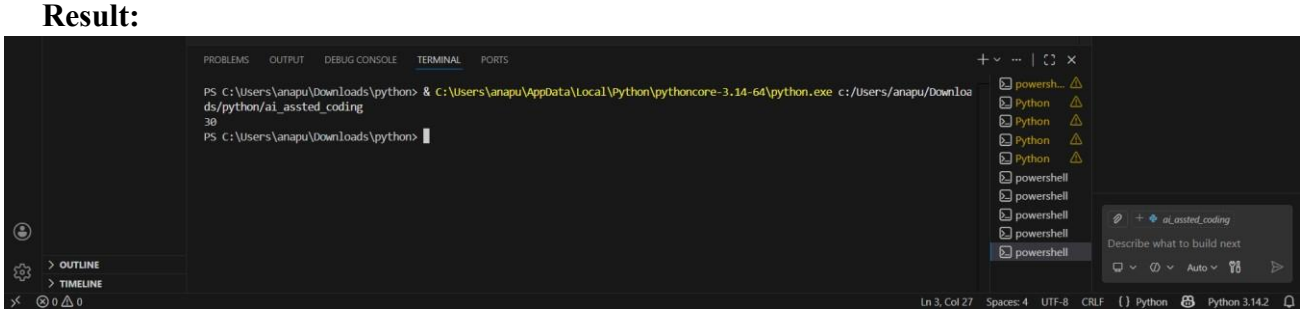
def f(a,b): return

a+b

Print(f(10,20)

Code & probmt

Result:



Observation:

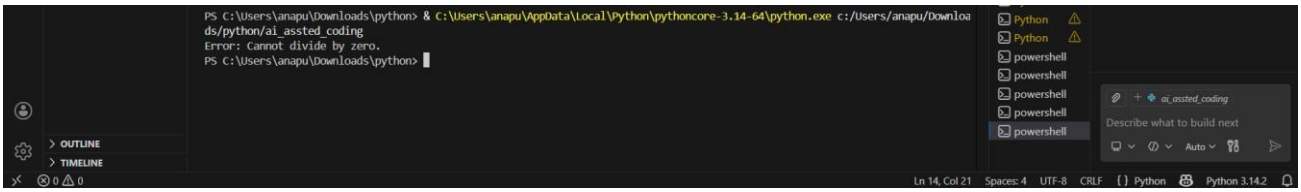
The original code used unclear and non-descriptive names, making it hard to understand the function's purpose. Using meaningful names improves clarity, readability, and overall code quality.

 # Task Description #2 – Missing Error Handling

Sample Input Code:
 # def divide(a, b):
 # return a / b
 # print(divide(10, 0))

Prompt &Code:

Result:



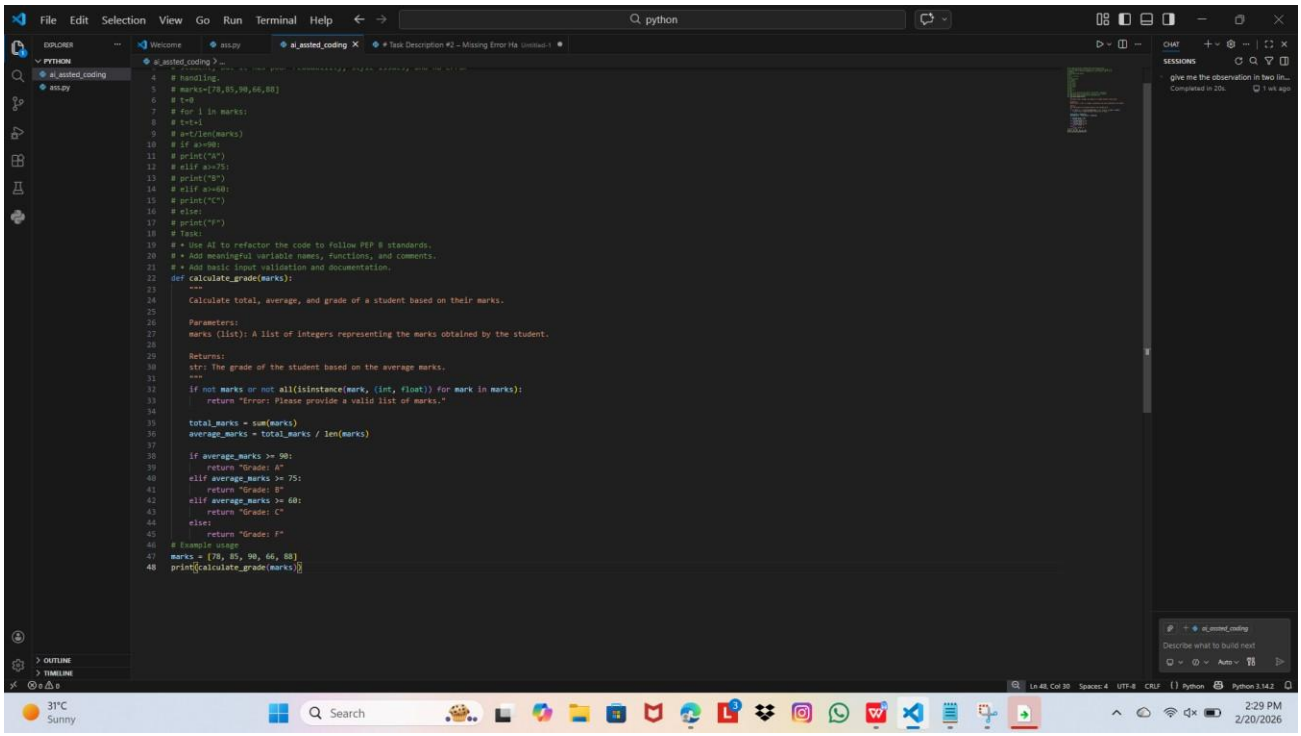
Observation:

The original code did not handle division by zero, which could cause the program to crash. Adding exception handling makes the program safer and provides a clear, user-friendly error message instead of stopping abruptly.

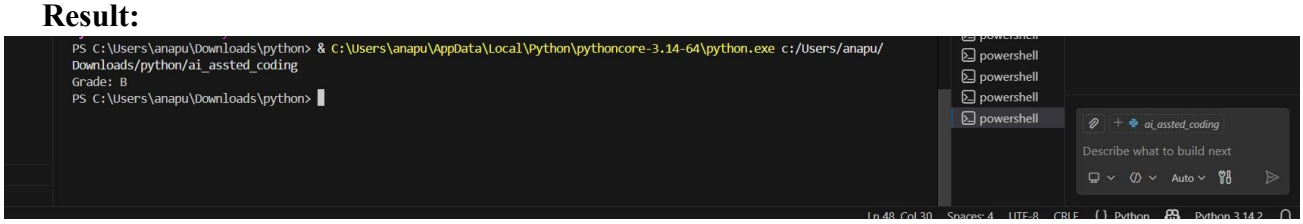
 # Task Description #3: Student Marks Processing System Sample
input code:

```
 marks=[78,85,90,66,88]
 t=0  for i in
marks:  t=t+i
a=t/len(marks)
if a>=90:
print("A")  elif
a>=75:
print("B")  elif
a>=60:
print("C")
else:
 print("F")
```

Prompt & Code:



Result:



Observation:

The refactored code improves readability by using meaningful variable names, proper formatting, functions, and clear documentation following PEP 8 standards.
It also adds basic input validation, making the program more reliable and user-friendly by preventing errors from invalid data.

 # Task Description #4: Use AI to add docstrings and inline comments

Sample input: def factorial(n):

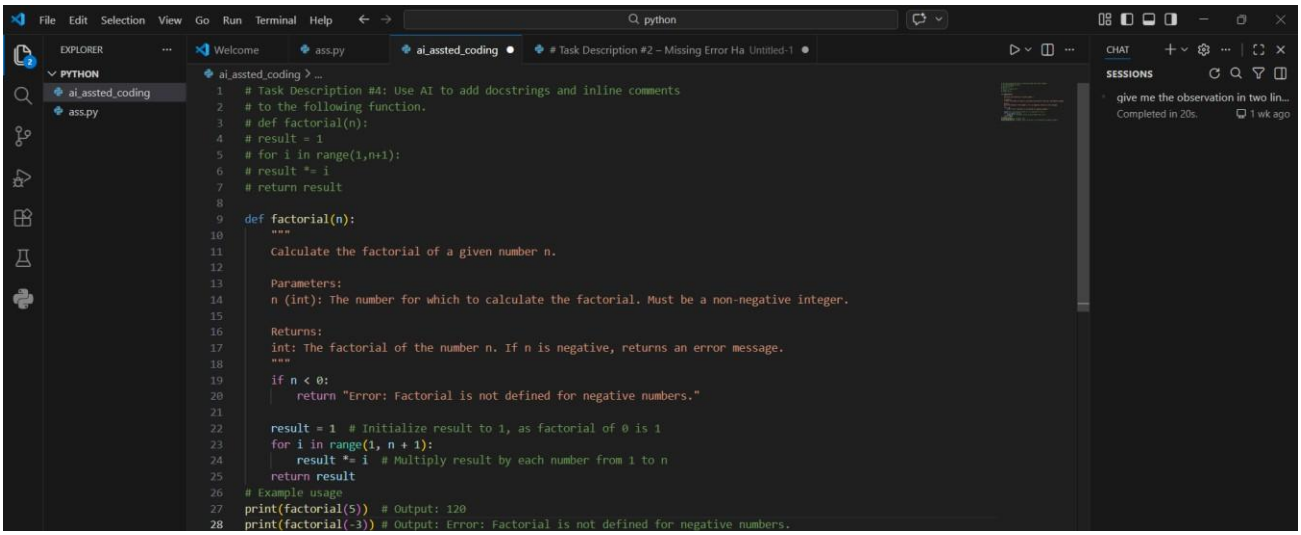
result = 1 for i in

range(1,n+1):

result *= i

return result

Prompt &Code:



```python
# Task Description #4: Use AI to add docstrings and inline comments
# to the following function.
# def factorial(n):
# result = 1
# for i in range(1,n+1):
# result *= i
# return result

def factorial(n):
    """
    Calculate the factorial of a given number n.

    Parameters:
    n (int): The number for which to calculate the factorial. Must be a non-negative integer.

    Returns:
    int: The factorial of the number n. If n is negative, returns an error message.
    """
    if n < 0:
        return "Error: Factorial is not defined for negative numbers."

    result = 1  # Initialize result to 1, as factorial of 0 is 1
    for i in range(1, n + 1):
        result *= i  # Multiply result by each number from 1 to n
    return result
# Example usage
print(factorial(5))  # Output: 120
print(factorial(-3)) # Output: Error: Factorial is not defined for negative numbers.
```

Result:

```
PS C:\Users\anapu\Downloads\python> & C:\Users\anapu\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/anapu/Downloads/python/ai_assted_coding
120
Error: Factorial is not defined for negative numbers.
PS C:\Users\anapu\Downloads\python>
```

Observation

The updated function includes a clear docstring and inline comments, which improve understanding of the code's purpose and logic.
It also adds basic validation for negative numbers, making the function more informative and reliable.

 # Task Description #5: Password Validation System (Enhanced)

Sample input:

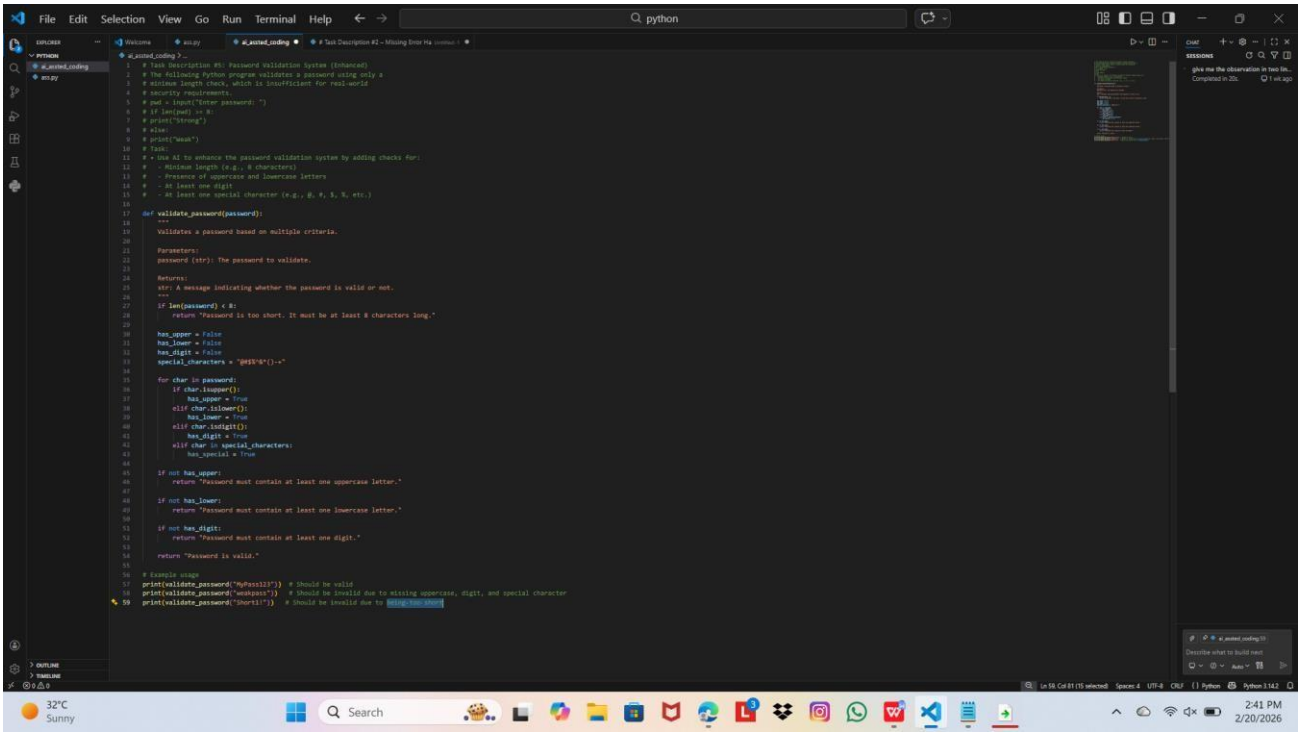
pwd = input("Enter password: ")

if len(pwd) >= 8:

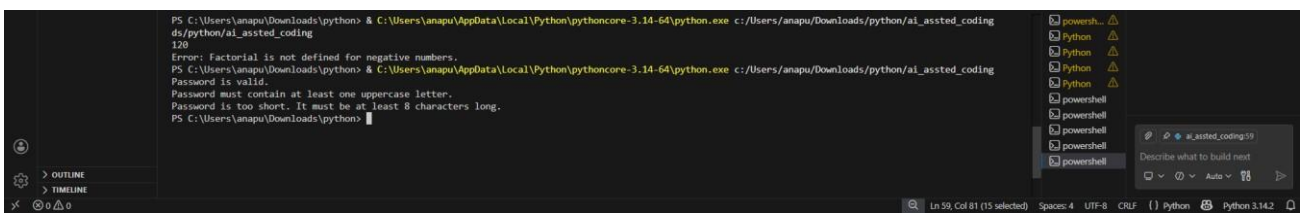
print("Strong") else:

print("Weak")

Prombt & Code:



Result:



Observation:

The enhanced password validation improves security by checking for length, uppercase and lowercase letters, digits, and special characters instead of just minimum length.
However, there is a small issue: the variable `has_special` is used without being initialized and the final check for a special character is missing, which could cause an error.